

Kai Zhu

PH. D. STUDENT · ZHEJIANG UNIVERSITY

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Summary

I am a Ph.D. student in Pharmaceutics at Zhejiang University, advised by Prof. Tingjun Hou. I am also a visiting student at the Italian Institute of Technology, working with Luigi Bonati and Prof. Michele Parrinello. My research focuses on developing machine learning-based enhanced sampling methods and their applications to atomic systems.

Education

College of Pharmaceutical Sciences, Zhejiang University

PH.D. IN MEDICINE CHEMISTRY

Hangzhou, China

Sep. 2023 – Jun. 2028

College of Pharmaceutical Sciences, Wenzhou Medical University

B.S. IN PHARMACEUTICAL SCIENCE

Wenzhou, China

Sep. 2019 – Jun. 2023

Research Experience

Italian Institute of Technology

RESEARCH INTERN, ADVISED BY PHD. LUIGI BONATI AND PROF. MICHELE PARRINELLO

Genova, Italy

Dec. 2025 – Dec. 2027

Research Topics: machine learning, enhanced sampling, atomic simulation

Zhejiang University

PHD STUDENT, ADVISED BY PROF. TINGJUN HOU AND PROF. DAN LI

Hangzhou, China

Jun. 2023 – Jun. 2025

Research Topics: machine learning, enhanced sampling, intrinsically disordered proteins

Honors & Awards

DOMESTIC

2025 Qiu Shi Flying Eagle Scholarship, Zhejiang University

Hangzhou, China

Selected Publications

† Co-first author; * Corresponding author;

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| 2026 | Targeting the Intrinsically Disordered AR-NTD through a Machine Learning-Based Enhanced Sampling Workflow
K. Zhu† , H. Wang†, J. Zhang†, R. Hu, L. Jiang, H. Zhang, Y. Kang, T. Hou*, D. Li*. Nat. Commun. (2026). | [4] |
| 2026 | A Scalable and Quantum-Accurate Foundation Model for Biomolecular Force Field via Linearly Tensorized Quadrangle Attention
Q. Su†, K. Zhu† , Q. Gou†, ..., J. Wang*, C. Hsieh*, T. Hou*. Nature Communications 17, 3639 (2026). | [3] |
| 2026 | Self-Supervised Evolution Operator Learning for High-Dimensional Dynamical Systems
G. Turri, L. Bonati, K. Zhu , M. Pontil, P. Novelli* ICLR 2026 . | [2] |
| 2025 | Enhanced Sampling in the Age of Machine Learning: Algorithms and Applications
K. Zhu† , E. Trizio†, J. Zhang, R. Hu, L. Jiang, T. Hou*, L. Bonati*. Chemical Reviews 126 (1), 671-713. | [1] |

Skills

Programming	Python, C++, C, $\LaTeX$, Linux
Frameworks	PyTorch, TensorFlow, NumPy, Git, Anaconda
Atomic simulation	GROMACS, PLUMED, AMBER, OpenMM, ASE, LAMMPS, CP2K